

## ONEGA, Russian Federation

WMO: 226410

Lat: 63.909N

Lon: 38.123E

Elev: 13

StdP: 101.17

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -29.9      | -26.3 | -32.5                         | 0.2 | -29.8 | -28.9 | 0.3 | -26.3 | 8.3                   | -6.1 | 7.2 | -4.3 | 0.7                   | 120  | 0.617 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.2                    | 27.6            | 19.4 | 25.3 | 18.1 | 23.3 | 17.2 | 20.5                | 25.8 | 19.2 | 24.0 | 18.0 | 22.1 | 2.6                  | 120  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.5                            | 13.4 | 23.3 | 17.3 | 12.4 | 22.0 | 16.1 | 11.5 | 20.9 | 59.1          | 25.8 | 54.9 | 24.1 | 50.9 | 22.1 | 25.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 7.7           | 6.4  | 5.8 | DB                         | -32.0 | 30.3               | 4.8 | 2.1                                                | -35.5 | 31.7       | -38.3 | 33.0       | -41.0 | 34.1       | -44.5 | 35.6 | (4) |
| (5)               |      |     | WB                         | -32.1 | 22.3               | 4.7 | 1.7                                                | -35.4 | 23.5       | -38.2 | 24.5       | -40.8 | 25.4       | -44.2 | 26.7 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 2.6         | -10.8 | -10.3 | -5.1  | 1.5   | 7.8   | 13.4  | 17.1  | 14.4  | 9.5   | 3.4   | -3.1  | -6.9 |  |  |
|                                                                        | DBStd       | 10.90       | 7.87  | 7.77  | 5.20  | 4.79  | 5.35  | 4.69  | 4.15  | 3.57  | 3.41  | 3.92  | 5.43  | 7.04 |  |  |
|                                                                        | HDD10.0     | 3246        | 645   | 567   | 469   | 259   | 109   | 18    | 2     | 3     | 49    | 206   | 394   | 522  |  |  |
|                                                                        | HDD18.3     | 5785        | 903   | 801   | 728   | 506   | 330   | 159   | 73    | 131   | 267   | 463   | 644   | 781  |  |  |
|                                                                        | CDD10.0     | 562         | 0     | 0     | 0     | 3     | 40    | 121   | 222   | 141   | 33    | 2     | 0     | 0    |  |  |
|                                                                        | CDD18.3     | 59          | 0     | 0     | 0     | 0     | 2     | 13    | 35    | 9     | 0     | 0     | 0     | 0    |  |  |
|                                                                        | CDH23.3     | 438         | 0     | 0     | 0     | 1     | 22    | 111   | 243   | 62    | 1     | 0     | 0     | 0    |  |  |
|                                                                        | CDH26.7     | 95          | 0     | 0     | 0     | 0     | 3     | 24    | 56    | 12    | 0     | 0     | 0     | 0    |  |  |
|                                                                        | WSAvg       | 2.4         | 2.2   | 2.3   | 2.5   | 2.3   | 2.6   | 2.4   | 2.2   | 2.0   | 2.3   | 2.6   | 2.5   | 2.6  |  |  |
|                                                                        | PrecAvg     | 591         | 36    | 29    | 29    | 31    | 38    | 58    | 63    | 74    | 70    | 67    | 57    | 43   |  |  |
| (15) Precipitation                                                     | PrecMax     | 756         | 66    | 62    | 66    | 75    | 74    | 137   | 123   | 150   | 140   | 125   | 100   | 86   |  |  |
|                                                                        | PrecMin     | 375         | 12    | 5     | 7     | 7     | 5     | 12    | 1     | 14    | 31    | 8     | 16    | 18   |  |  |
|                                                                        | PrecStd     | 77          | 14    | 15    | 14    | 17    | 15    | 32    | 29    | 32    | 28    | 24    | 21    | 16   |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |      |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 2.7   | 3.6   | 7.8   | 19.4  | 26.4  | 29.4  | 30.7  | 28.5  | 21.5  | 13.9  | 8.1   | 5.0  |  |  |
|                                                                        |             | MCWB        | 1.6   | 2.2   | 3.6   | 10.6  | 15.9  | 20.5  | 20.8  | 20.2  | 16.1  | 11.5  | 6.7   | 3.9  |  |  |
|                                                                        | 2%          | DB          | 1.3   | 2.1   | 4.3   | 13.7  | 22.1  | 26.8  | 28.1  | 24.9  | 18.1  | 11.6  | 6.0   | 2.7  |  |  |
|                                                                        |             | MCWB        | 0.4   | 1.0   | 1.4   | 8.6   | 13.8  | 18.2  | 20.2  | 18.2  | 14.0  | 10.1  | 5.1   | 1.8  |  |  |
|                                                                        | 5%          | DB          | 0.4   | 0.9   | 2.9   | 10.2  | 19.4  | 24.0  | 26.1  | 22.3  | 16.2  | 10.2  | 4.4   | 1.9  |  |  |
|                                                                        |             | MCWB        | -0.2  | 0.0   | 0.9   | 6.1   | 12.6  | 16.5  | 19.2  | 17.0  | 13.1  | 9.0   | 3.7   | 1.2  |  |  |
|                                                                        | 10%         | DB          | -0.8  | -0.4  | 1.9   | 7.9   | 16.6  | 21.6  | 24.0  | 20.4  | 14.7  | 8.7   | 3.1   | 1.0  |  |  |
|                                                                        |             | MCWB        | -1.4  | -1.2  | 0.3   | 4.4   | 11.1  | 15.3  | 18.1  | 16.1  | 12.2  | 7.6   | 2.3   | 0.3  |  |  |
|                                                                        | 0.4%        | WB          | 1.6   | 2.4   | 3.9   | 12.0  | 17.2  | 21.7  | 22.5  | 21.1  | 16.7  | 12.1  | 7.0   | 4.3  |  |  |
|                                                                        |             | MCDB        | 2.3   | 3.3   | 6.8   | 18.2  | 23.2  | 27.3  | 28.1  | 26.4  | 20.2  | 13.2  | 8.0   | 4.9  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB          | 0.6   | 1.1   | 2.1   | 8.8   | 15.0  | 19.3  | 21.1  | 19.2  | 14.7  | 10.5  | 5.1   | 2.0  |  |  |
|                                                                        |             | MCDB        | 1.2   | 1.9   | 3.5   | 13.6  | 20.1  | 25.4  | 26.4  | 23.3  | 17.2  | 11.5  | 5.9   | 2.5  |  |  |
|                                                                        | 5%          | WB          | -0.2  | 0.2   | 1.2   | 6.5   | 13.2  | 17.6  | 19.9  | 17.8  | 13.5  | 9.1   | 3.8   | 1.3  |  |  |
|                                                                        |             | MCDB        | 0.3   | 0.9   | 2.5   | 9.8   | 18.4  | 22.3  | 25.0  | 21.4  | 15.7  | 10.1  | 4.4   | 1.8  |  |  |
|                                                                        | 10%         | WB          | -1.4  | -1.2  | 0.4   | 4.7   | 11.6  | 15.8  | 18.8  | 16.6  | 12.4  | 7.6   | 2.4   | 0.4  |  |  |
|                                                                        |             | MCDB        | -0.8  | -0.5  | 1.8   | 7.4   | 16.2  | 20.7  | 23.1  | 19.8  | 14.5  | 8.6   | 2.9   | 0.9  |  |  |
|                                                                        | 5% DB       | MDBR        | 5.7   | 6.4   | 7.5   | 7.5   | 8.4   | 8.5   | 8.2   | 7.5   | 6.3   | 4.0   | 4.0   | 4.8  |  |  |
|                                                                        |             | MCDBR       | 4.8   | 4.9   | 6.9   | 10.5  | 12.9  | 11.5  | 10.9  | 10.4  | 8.7   | 5.2   | 3.9   | 3.6  |  |  |
|                                                                        |             | MCWBR       | 4.6   | 4.5   | 5.2   | 6.5   | 6.9   | 5.7   | 5.3   | 5.3   | 5.4   | 4.1   | 3.8   | 3.3  |  |  |
|                                                                        |             | MCWBR       | 4.7   | 4.9   | 5.7   | 10.1  | 11.5  | 9.8   | 9.6   | 8.9   | 7.5   | 4.9   | 4.0   | 3.5  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR       | 4.5   | 4.6   | 4.5   | 6.5   | 6.8   | 5.7   | 5.1   | 5.4   | 5.1   | 4.2   | 3.8   | 3.4  |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |      |  |  |
|                                                                        | taub        |             | 0.232 | 0.288 | 0.316 | 0.353 | 0.357 | 0.351 | 0.387 | 0.364 | 0.336 | 0.311 | 0.369 | N/A  |  |  |
|                                                                        |             | taud        | 1.979 | 2.089 | 2.134 | 2.231 | 2.413 | 2.468 | 2.394 | 2.437 | 2.500 | 2.351 | 2.607 | N/A  |  |  |
| (42) Clear-Sky Solar Irradiance                                        | Ebn at Noon |             | 329   | 606   | 758   | 817   | 855   | 869   | 819   | 807   | 758   | 600   | 316   | N/A  |  |  |
|                                                                        |             | Edh at Noon | 29    | 65    | 95    | 109   | 100   | 96    | 102   | 89    | 68    | 51    | 26    | N/A  |  |  |
|                                                                        | RadAvg      |             | 0.15  | 0.69  | 2.08  | 3.70  | 4.73  | 5.43  | 5.07  | 3.60  | 2.01  | 0.73  | 0.17  | 0.06 |  |  |
| (45) All-Sky Solar Radiation                                           | RadStd      |             | 0.02  | 0.13  | 0.25  | 0.32  | 0.46  | 0.51  | 0.55  | 0.40  | 0.23  | 0.10  | 0.03  | 0.01 |  |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |  |  |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|--|--|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |  |  |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |  |  |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | +0.79 | +0.98 | N/A         | N/A     | N/A     | N/A     |  |  |
| (47) Regional (5 neighbors) |  | +0.71 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -208    | N/A     | N/A     |  |  |

Nomenclature: See separate page